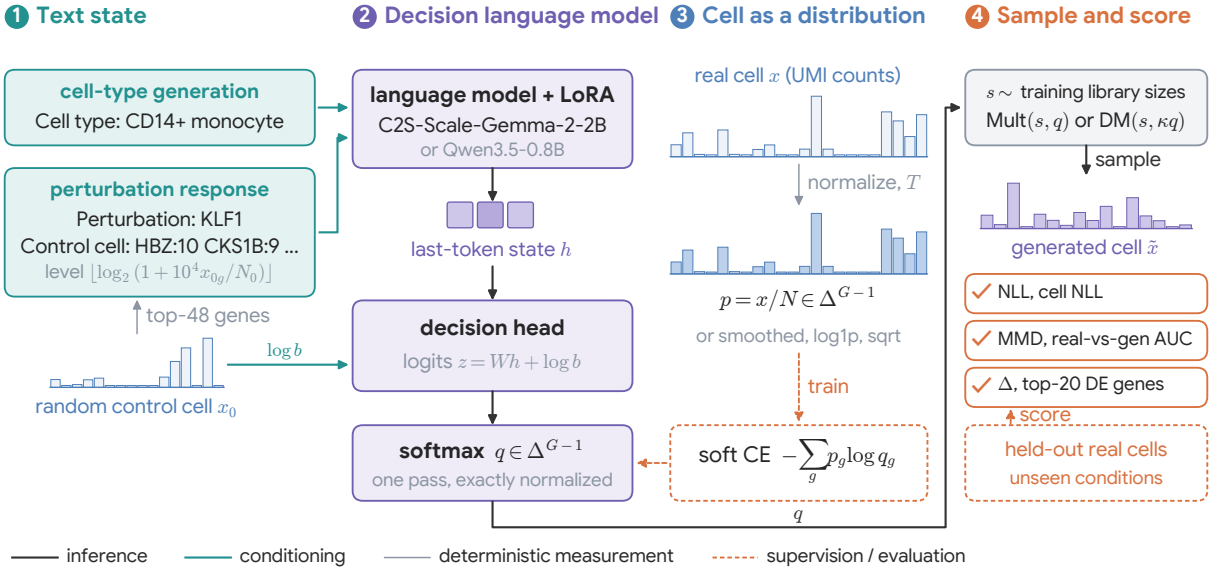
1 Text state
2 Decision language model
3 Cell as a distribution
4 Sample and score
cell-type generation
Cell type: CD14+ monocyte
perturbation response
Perturbation: KLF1
Control cell: HBZ:10 CKS1B:9 ...
level $\lfloor \log_2(1+10^4 x_{0g}/N_0) \rfloor$
top-48 genes
$\log b$
random control cell x_0
language model + LoRA
C2S-Scale-Gemma-2-2B
or Qwen3.5-0.8B
last-token state h
decision head
logits $z = Wh + \log b$
softmax $q \in \Delta^{G-1}$
one pass, exactly normalized
real cell x (UMI counts)
normalize, T
$p = x/N \in \Delta^{G-1}$
or smoothed, log1p, sqrt
train
soft CE $-\sum_g p_g \log q_g$
q
$s \sim$ training library sizes
Mult(s, q) or DM$(s, \kappa q)$
sample
generated cell $\tilde{x}$
NLL, cell NLL
MMD, real-vs-gen AUC
Δ, top-20 DE genes
score
held-out real cells
unseen conditions
inference
conditioning
deterministic measurement
supervision / evaluation